

Moller FADC DAQ

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Introduction

- The new Moller DAQ based on FADC unit has been used during PREX experiment. FADC DAQ was running in parallel with old DAQ.
- The main goal of FADC DAQ is to reduce systematic errors by:
 - measuring the asymmetries using all the event data
 - reducing deadtime corrections due to FADC pipeline implementation;
 - analyzing the pile-up and other effects with “Data Trigger”.

FADC DAQ Parameters

- 16 Analog Channels
- 250MHz sample rate
- 12 bits/sample resolution
- On board of FPGA coincidence logic, scalers unit and logic to generate triggers
- Threshold levels, time delays, pulse width, trigger prescales are all software controlled

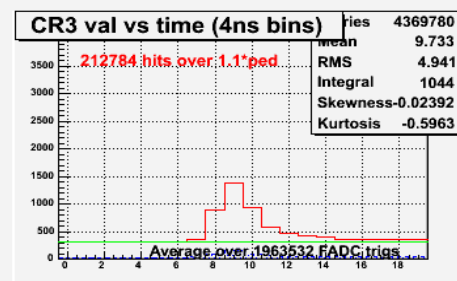
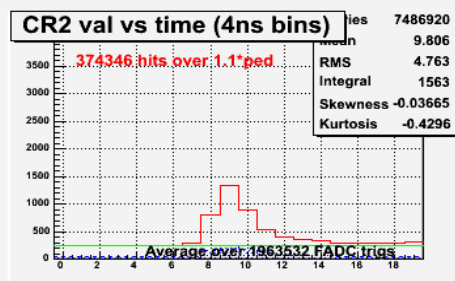
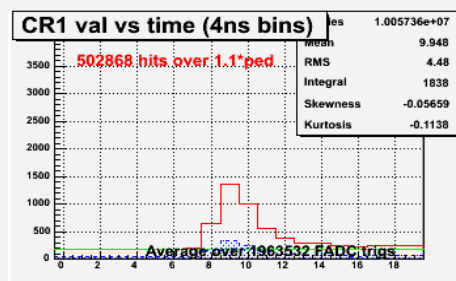
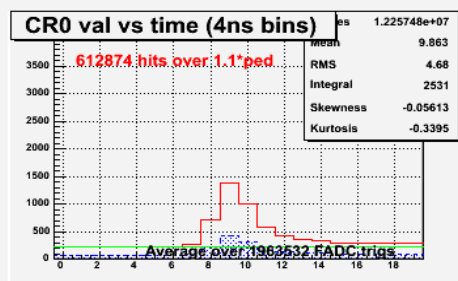
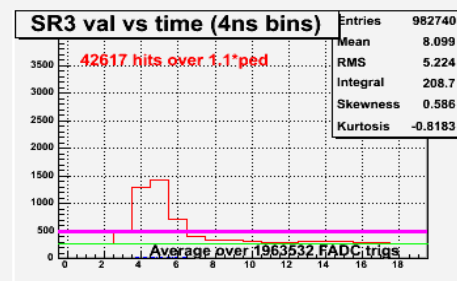
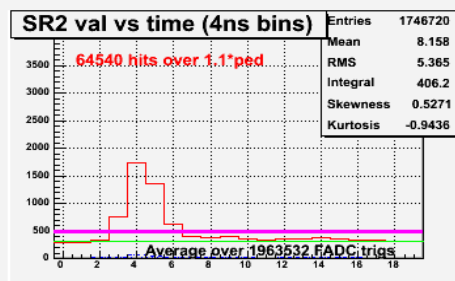
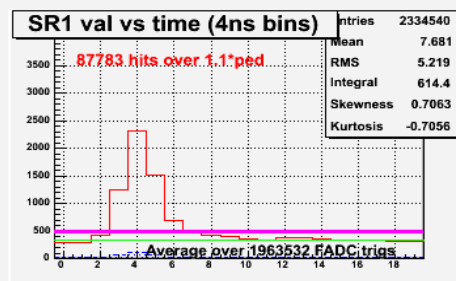
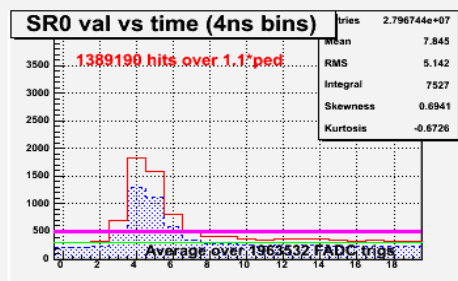
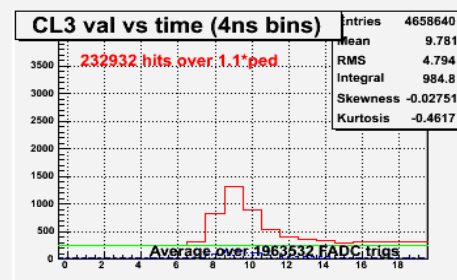
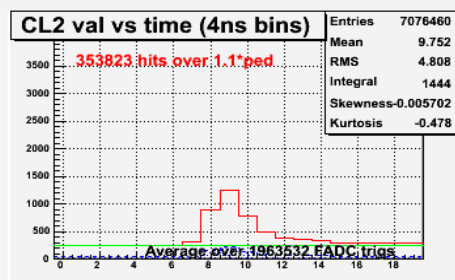
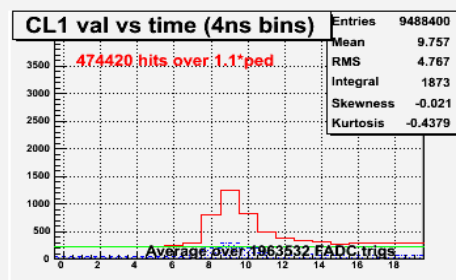
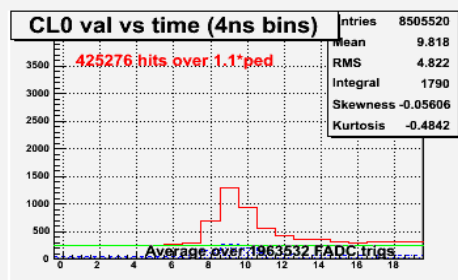
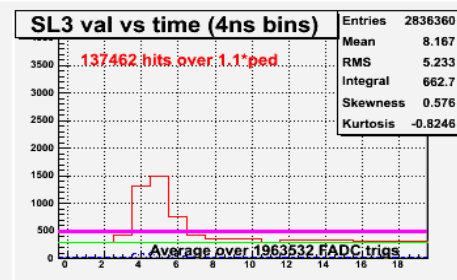
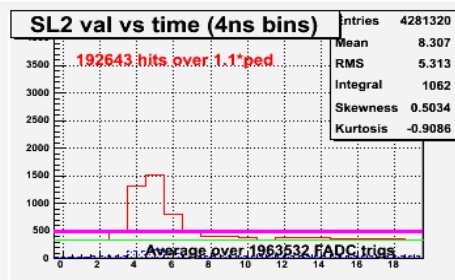
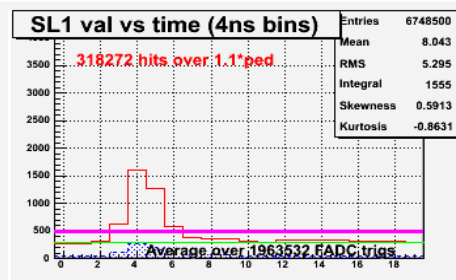
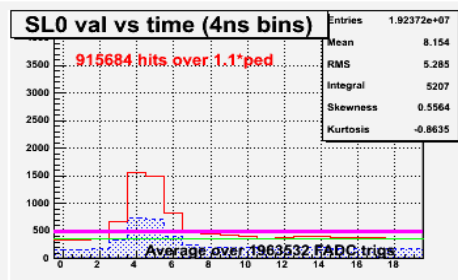
FADC DAQ Parameters

- New Moller DAQ computer hamoller.jlab.org
 - 4 CPU cores
 - 2TB disk storage
 - Fast Gbit network
 - Data stream rate up to 50MB/s
 - Shares “adaqfs” file system
- FADC DAQ has ROOT based analyzer

FADC DAQ Parameters

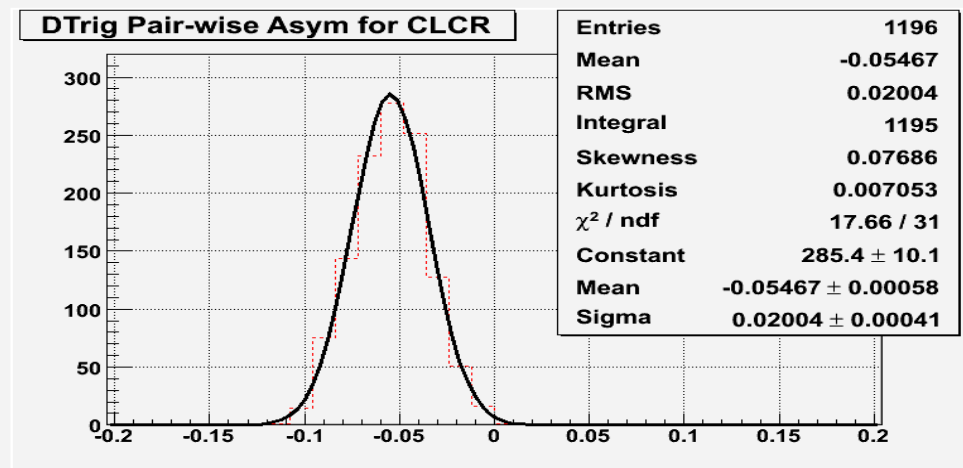
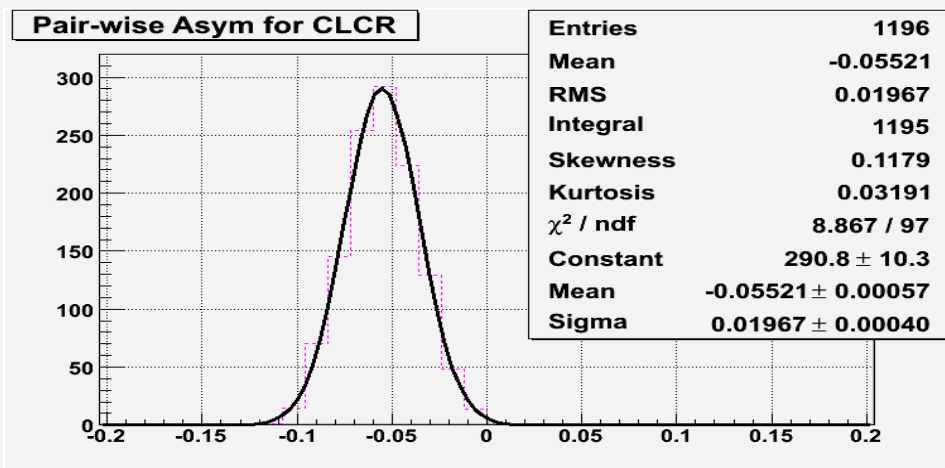
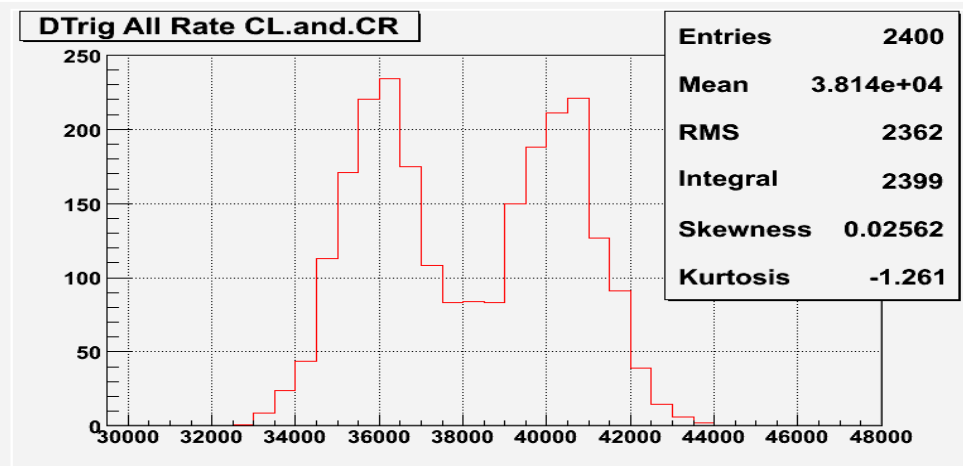
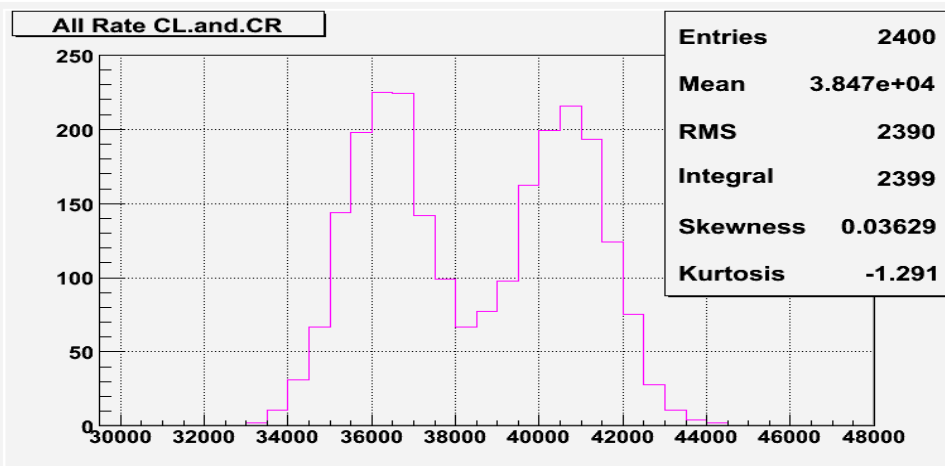
- Auxiliary support modules to FADC
 - CAEN V560 scaler (16 channel, 100MHz)
BCM, moller target position, 100kHz clock
 - CAEN V792 QDC (32 channels) records
MPS, QRT and Helicity signal states
- FADC DAQ generates two type of triggers:
 - Helicity flipping triggers
 - Data triggers

Example of Events from Data Trigger



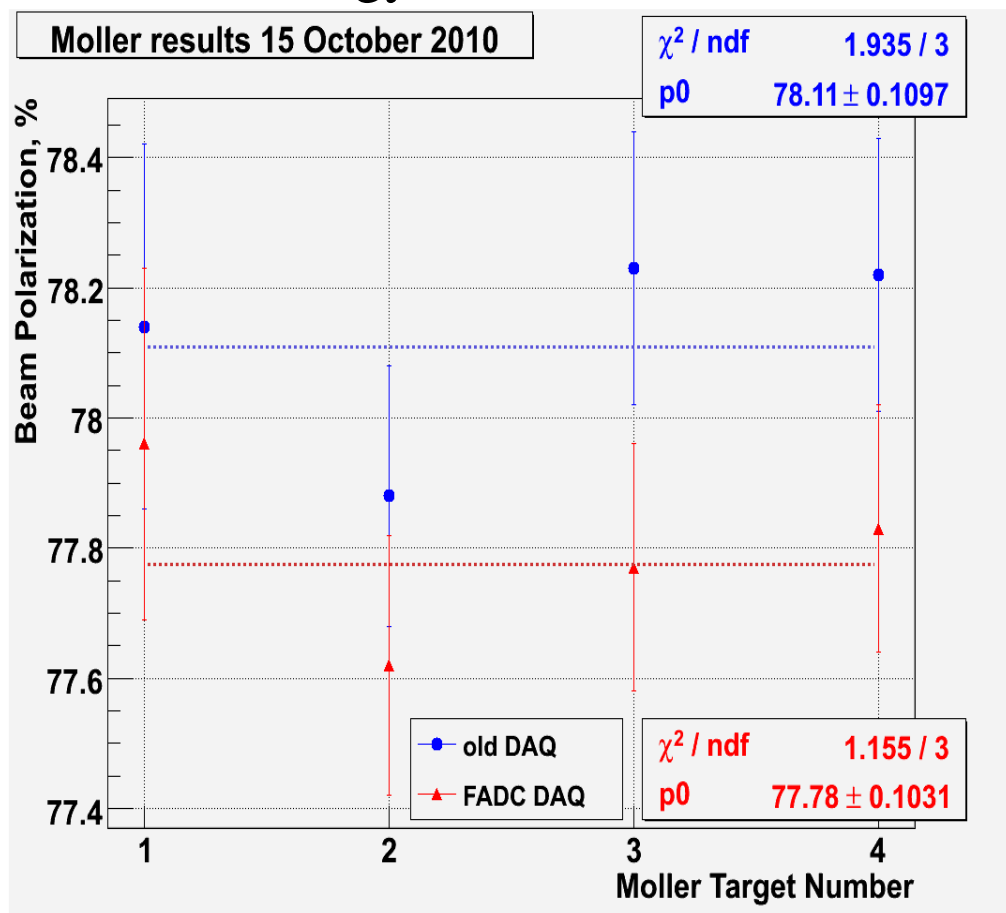
FADC Scalers Rate vs Data Trigger Rate

Run #2046, targ#3, 1.2uA	Coinc Rate CL and CR	Asymmetry CL and CR
FADC Scalers	38470 +/-2390	-0.05521 +/-0.01967
FADC DTrig	38140 +/-2362	-0.05467 +/-0.0200
Old DAQ	32697 +/-1021	-0.0564 +/-0.0220

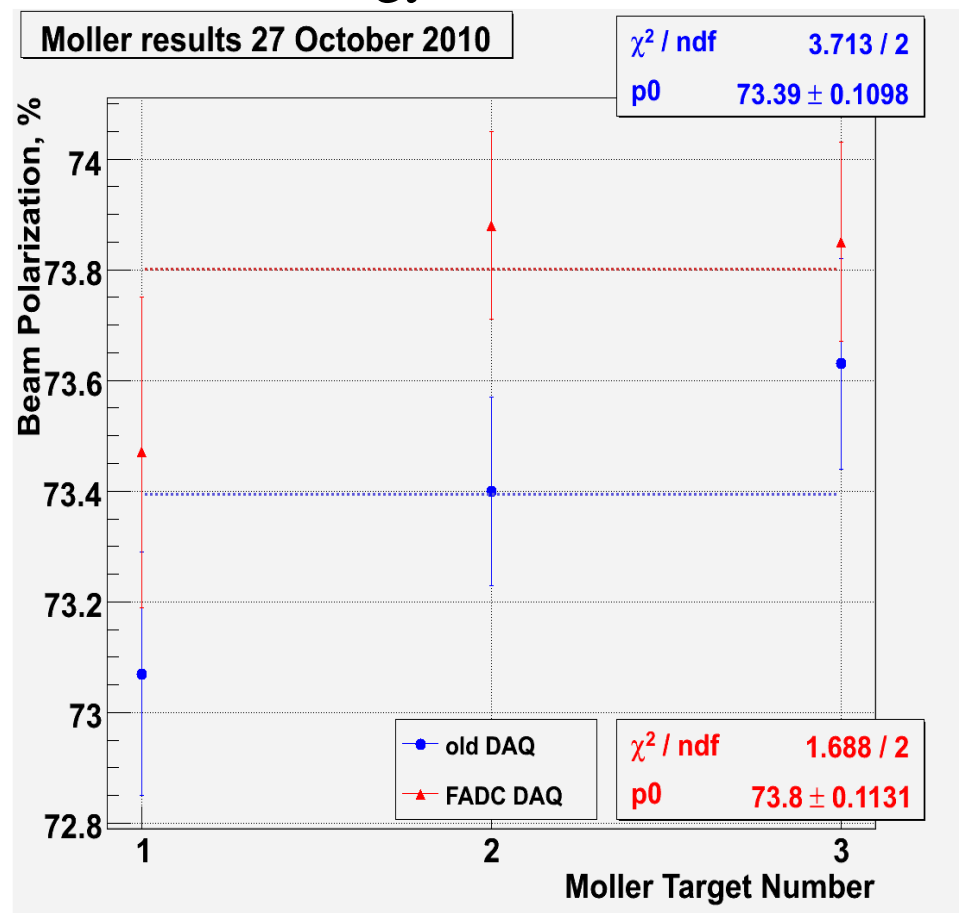


Beam Polarization Measurements

Beam energy 5548.91MeV



Beam energy 4455.7MeV



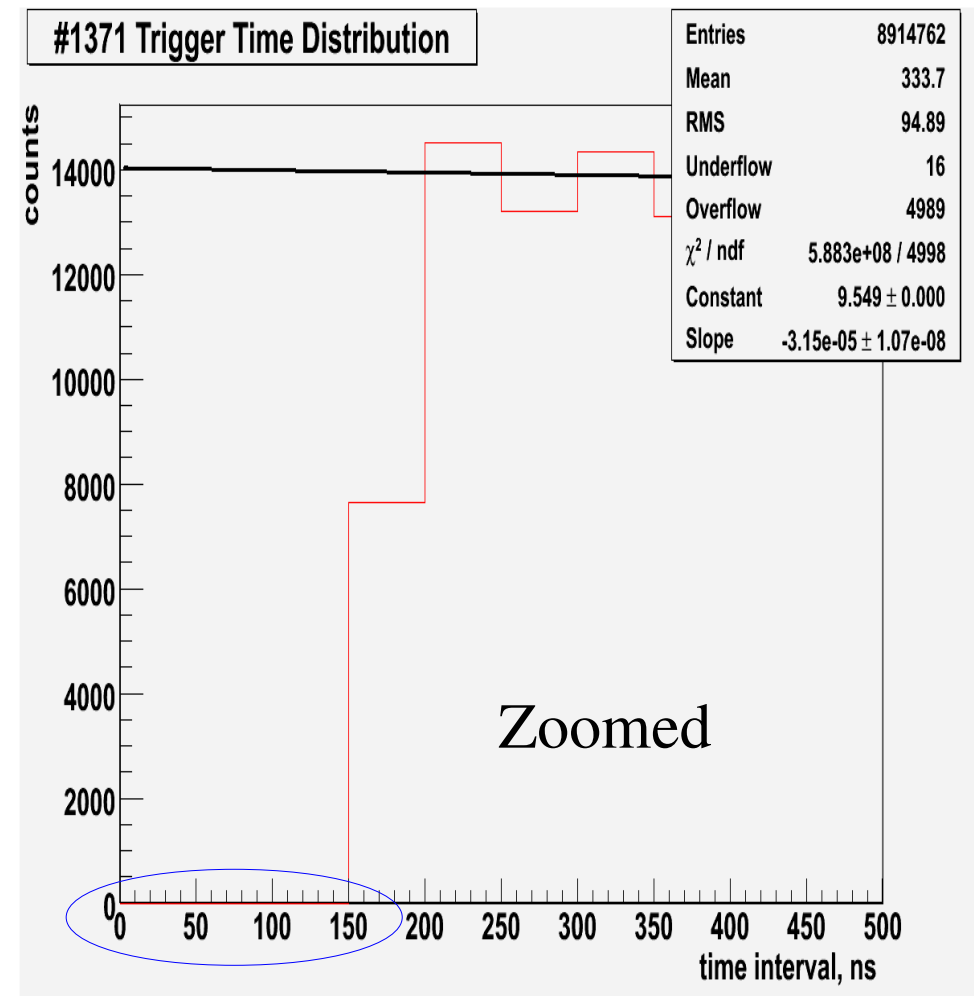
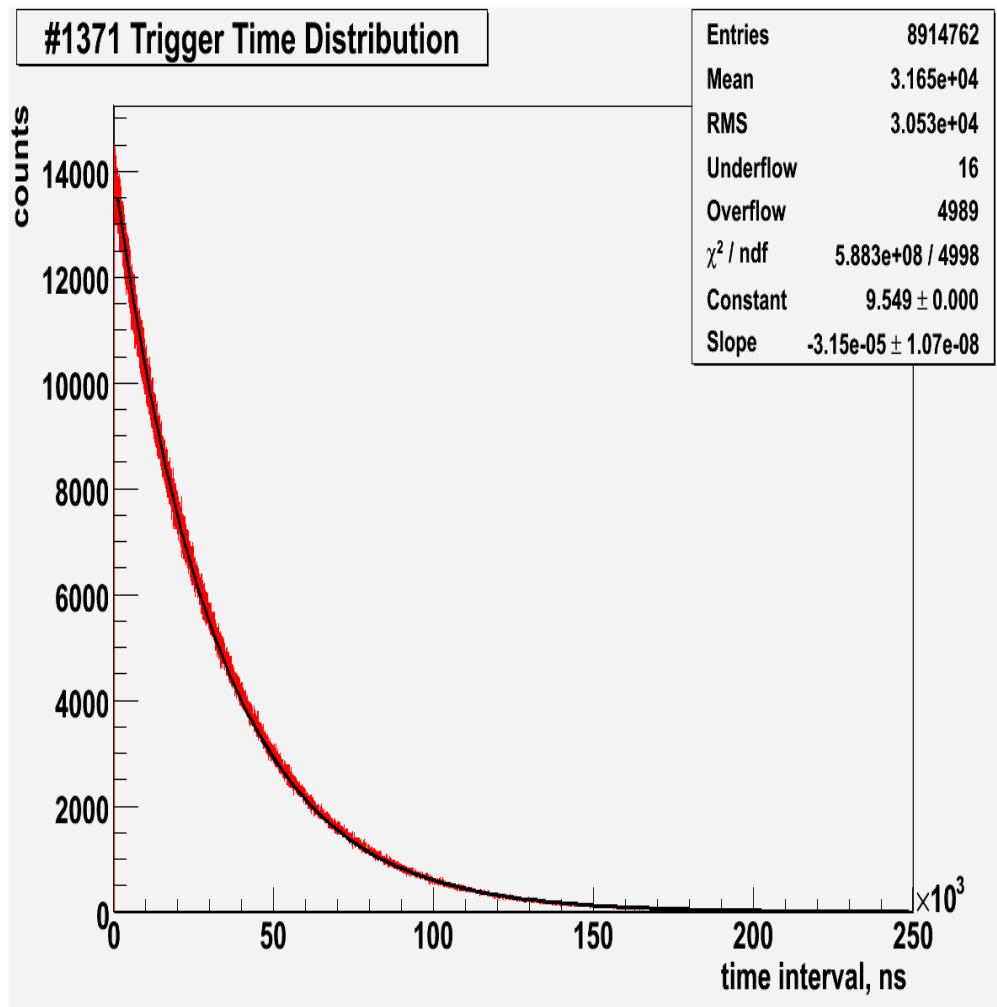
Example of rates of **old DAQ** (blue) and **FADC DAQ** (red) : target #1 @3.0uA

run	Left	Right	Coinc.	Accid.	BCM	Raw.Asymm
14903	49885	64565	24535	61.	45048.	0.0447+/- 0.0005
2383	56593	66088	28331	79.	42462	0.0446+/- 0.0004

Data Trigger dead-time

Distribution of time intervals between Data triggers.

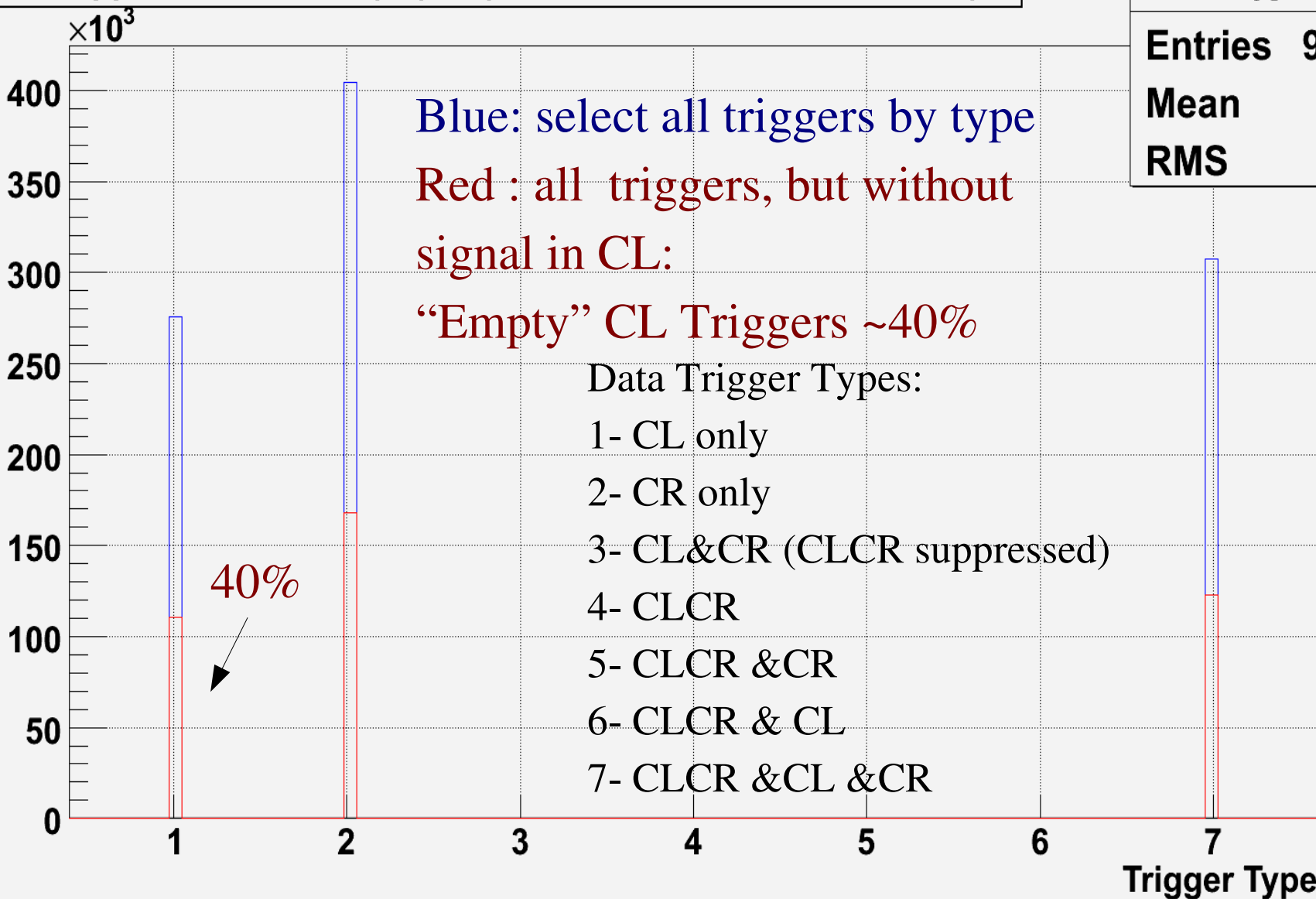
This is the Poisson probability for no events and it is an exponential function. Estimated dead-time is $\sim 150\text{ns}$.



Data Triggers by Type

Problem: “Empty” Triggers

Data Trigger Distribution by Type (Run #2387 CL:CR:CLCR = 1:1:1)



Conclusion

- FADC DAQ is fully operational
- Events rate is about $\sim 10\%$ - 20% higher than the old DAQ due to different thresholds settings
- Asymmetry is consistent with old DAQ, difference remains under 0.6%
- Did not have precisely deadtime measurements with beam
- Problem of “Empty” data triggers is under investigation

Things to Do

- Precisely dead-time measurements
- Improvement FADC analyzer to produce reliable results and analysis of data trigger: to estimate Levchuk and pile-up effects
- Increasing of DAQ through pass and data writing to disks is required
- Detailed documentation of FADC DAQ and how to analyze data

FADC DAQ

- Thanks to E. Chudakov, R. Michaels,
O.Glamazdin, Brad Sawatzky, Ahmed Zafar and
Jixie Zhang